



```

name: <unnamed>
log: C:\Users\juani\OneDrive\Documents\UAH\PhD Documents\INEDatos\Analysis\Tut
> orialLog.smcl
log type: smcl
opened on: 9 Mar 2026, 12:21:43

```

```

1 .
2 . * Describe data
3 . describe

```

Contains data from **dt.dta**

Observations: **19,503**

Variables: **17**

08 Mar 2026 19:26

Variable name	Storage type	Display format	Value label	Variable label
factor2	double	%10.0g		
sexo	long	%12.0g	sexo	
edad_cat3	long	%12.0g	edad_cat3	
clase_2	long	%12.0g	clase_2	
survey	long	%12.0g	survey	
survey2	long	%12.0g	survey2	
sedentarismo	long	%12.0g	sedentarismo	
urb_rur	long	%12.0g	urb_rur	
sexo_num	double	%10.0g		
edad_cat3_num	double	%10.0g		
clase_2_num	double	%10.0g		
urb_rur_num	double	%10.0g		
survey_num	double	%10.0g		
survey2_num	double	%10.0g		
sedentarismo2	double	%10.0g		
stratum	long	%12.0g	stratum	
strataN	long	%12.0g		

Sorted by:

```
4 . summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
factor2	19,503	14.54011	11.2549	1	100
sexo	19,503	1.486592	.499833	1	2
edad_cat3	19,503	1.954571	.8247576	1	3
clase_2	19,503	1.568733	.4952659	1	2
survey	19,503	2.72153	1.350954	1	5
survey2	19,503	1.67041	.4700765	1	2
sedentarismo	19,503	1.141004	.3480345	1	2
urb_rur	19,503	2.243757	.8042797	1	3
sexo_num	19,503	1.486592	.499833	1	2
edad_cat3_num	19,503	1.954571	.8247576	1	3
clase_2_num	19,503	1.568733	.4952659	1	2
urb_rur_num	19,503	2.243757	.8042797	1	3
survey_num	19,503	2.72153	1.350954	1	5
survey2_num	19,503	1.67041	.4700765	1	2
sedentaris~2	19,503	.1410039	.3480345	0	1
stratum	19,503	36.54248	20.77069	1	72
strataN	19,503	338.2991	135.8224	62	560

```

5 .
6 . **# Model 0
7 . * Fit the two-level logistic regression with no covariates
8 . melogit sedentarisimo2 || stratum:, or baselevels

```

Fitting fixed-effects model:

```

Iteration 0: Log likelihood = -7937.9034
Iteration 1: Log likelihood = -7933.4667
Iteration 2: Log likelihood = -7933.4648
Iteration 3: Log likelihood = -7933.4648

```

Refining starting values:

```

Grid node 0: Log likelihood = -7720.0617

```

Fitting full model:

```

Iteration 0: Log likelihood = -7720.0617 (not concave)
Iteration 1: Log likelihood = -7709.5751
Iteration 2: Log likelihood = -7703.2899
Iteration 3: Log likelihood = -7702.5497
Iteration 4: Log likelihood = -7702.5415
Iteration 5: Log likelihood = -7702.5415

```

```

Mixed-effects logistic regression      Number of obs      =      19,503
Group variable: stratum              Number of groups   =         72

Obs per group:
      min =         62
      avg =      270.9
      max =        560

```

```

Integration method: mvaghermite      Integration pts.   =         7

```

```

Log likelihood = -7702.5415            Wald chi2(0)       =         .
                                      Prob > chi2         =         .

```

sedentaris~2	Odds	Std. err.	z	P> z	[95% conf. interval]	
_cons	0.14	0.01	-29.10	0.000	0.12	0.16
stratum var(_cons)	0.28	0.06			0.19	0.41

Note: Estimates are transformed only in the first equation to odds.

```

LR test vs. logistic model: chibar2(01) = 461.85      Prob >= chibar2 = 0.0000

```

```

9 .
10. * Save the model results
11. estimates save "m0.ster", replace
    file m0.ster saved

12.
13. * Store the between-stratum variance
14. scalar m0sigma2 = _b[/var(_cons[stratum])]

15.
16. * Predict the fitted linear predictor
17. predict m0_total, eta
    (predictions based on fixed effects and posterior means of random effects)
    (using 7 quadrature points)

```

```

18.
19. * Predict the linear predictor for the fixed portion of the model only
20. predict m0_fixed, xb

21.
22. **# Model 1
23. * Fit the two-level logistic regression with covariates
24. melogit sedentarismo2 i.sexo i.edad_cat3 i.clase_2 i.urb_rur i.survey2 || stratum:,
> ///
> or baselevels

```

Fitting fixed-effects model:

```

Iteration 0: Log likelihood = -7715.2179
Iteration 1: Log likelihood = -7656.5568
Iteration 2: Log likelihood = -7656.429
Iteration 3: Log likelihood = -7656.429

```

Refining starting values:

```
Grid node 0: Log likelihood = -7709.3085
```

Fitting full model:

```

Iteration 0: Log likelihood = -7709.3085 (not concave)
Iteration 1: Log likelihood = -7681.4963 (not concave)
Iteration 2: Log likelihood = -7654.7224
Iteration 3: Log likelihood = -7649.6464
Iteration 4: Log likelihood = -7649.4416
Iteration 5: Log likelihood = -7649.4122
Iteration 6: Log likelihood = -7649.4122

```

Mixed-effects logistic regression
Group variable: **stratum**

```

Number of obs    =    19,503
Number of groups =         72

```

```

Obs per group:
    min =         62
    avg =       270.9
    max =         560

```

Integration method: **mvaghermite**

```
Integration pts. =         7
```

Log likelihood = **-7649.4122**

```

Wald chi2(7)     =    252.56
Prob > chi2      =    0.0000

```

sedentarismo2	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
sexo						
Male	1.00	(base)				
Female	1.91	0.12	10.62	0.000	1.69	2.15
edad_cat3						
6-9	1.00	(base)				
10-12	1.05	0.08	0.65	0.518	0.91	1.22
13-15	1.64	0.12	6.84	0.000	1.42	1.89
clase_2						
Non-Manual W..	1.00	(base)				
Manual Workers	1.62	0.10	7.80	0.000	1.43	1.82
urb_rur						
Rural	1.00	(base)				
Semi-urban	1.34	0.11	3.64	0.000	1.14	1.57
Urban	1.36	0.10	4.02	0.000	1.17	1.58
survey2						
Post	1.00	(base)				
Pre	1.32	0.08	4.39	0.000	1.17	1.49
_cons	0.05	0.00	-30.01	0.000	0.04	0.06
stratum						

var(_cons)	0.03	0.01	0.01	0.06
------------	------	------	------	------

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

LR test vs. logistic model: chibar2(01) = 14.03 Prob >= chibar2 = 0.0001

```

25.
26. * Save the model results
27. estimates save "ml.ster", replace
    file ml.ster saved

28.
29. * Store the between-stratum variance
30. scalar mlsigma2 = _b[/var(_cons[stratum])]

31.
32. * Predict the fitted linear predictor
33. predict ml_total, eta
    (predictions based on fixed effects and posterior means of random effects)
    (using 7 quadrature points)

34.
35. * Predict the linear predictor for the fixed portion of the model only
36. predict ml_fixed, xb

37.
38. * Predict the standard error of the fixed-portion linear prediction
39. predict ml_fixed_se, stdp

40.
41. * Predict the stratum random effect and its standard error
42. predict ml_random, reffect reses(ml_random se)
    (calculating posterior means of random effects)
    (using 7 quadrature points)

43.
44. **# Save data at individual-level
45.
46. * Compress the data
47. compress
    variable sexo was long now byte
    variable edad_cat3 was long now byte
    variable clase_2 was long now byte
    variable survey was long now byte
    variable survey2 was long now byte
    variable sedentarismo was long now byte
    variable urb_rur was long now byte
    variable stratum was long now byte
    variable strataN was long now int
    variable sexo_num was double now byte
    variable edad_cat3_num was double now byte
    variable clase_2_num was double now byte
    variable urb_rur_num was double now byte
    variable survey_num was double now byte
    variable survey2_num was double now byte
    variable sedentarismo2 was double now byte
    (1,462,725 bytes saved)

48.
49. * Save the data

```

```

50. save "individual.dta", replace
    file individual.dta saved

51.
52. **# Save data at stratum-level
53.
54. * Collapse the data down to a stratum-level dataset
55. collapse (count) n = sedentarismo2 (mean) sedentarismo2, ///
    >     by(stratum sexo edad_cat3 clase_2 urb_rur survey2 ///
    >         m0_total m0_fixed m1_total m1_fixed m1_fixed_se m1_random_se m1_random)

56.
57. * Move the list of variables to the beginning of the dataset
58. order stratum sexo edad_cat3 clase_2 urb_rur survey2 n sedentarismo2

59.
60. * Convert the outcome from a proportion to a percentage
61. replace sedentarismo2 = 100*sedentarismo2
    (72 real changes made)

62.
63. * Set the display format to 1dp
64. format %9.1f sedentarismo2

65.
66. * Set the display format to 2dp
67. format %9.2f sedentarismo2 ///
    >     m0_total m0_fixed m1_total m1_fixed m1_fixed_se m1_random_se m1_random

68.
69. * Compress the data
70. compress
    variable n was long now int
    (144 bytes saved)

71.
72. * Save the data
73. save "stratum.dta", replace
    file stratum.dta saved

74.
75. **# Individual-level descriptive statistics
76.
77. * Load the individual-level data
78. use "individual.dta", clear

79.
80. * Total sample size
81. codebook stratum, compact

```

Variable	Obs	Unique	Mean	Min	Max	Label
stratum	19503	72	36.54248	1	72	

```

82.
83. * Tabulate each individual characteristic
84. tabulate sexo

```

sexo	Freq.	Percent	Cum.
Male	10,013	51.34	51.34
Female	9,490	48.66	100.00
Total	19,503	100.00	

85. tabulate edad_cat3

edad_cat3	Freq.	Percent	Cum.
6-9	7,096	36.38	36.38
10-12	6,197	31.77	68.16
13-15	6,210	31.84	100.00
Total	19,503	100.00	

86. tabulate clase_2

clase_2	Freq.	Percent	Cum.
Non-Manual Workers	8,411	43.13	43.13
Manual Workers	11,092	56.87	100.00
Total	19,503	100.00	

87. tabulate urb_rur

urb_rur	Freq.	Percent	Cum.
Rural	4,510	23.12	23.12
Semi-urban	5,729	29.37	52.50
Urban	9,264	47.50	100.00
Total	19,503	100.00	

88. tabulate survey2

survey2	Freq.	Percent	Cum.
Post	6,428	32.96	32.96
Pre	13,075	67.04	100.00
Total	19,503	100.00	

89.

90. * Summarize the individual outcome

91. tabstat sedentarismo2, statistics(n mean sd min max) format(%3.1f)

Variable	N	Mean	SD	Min	Max
sedentaris~2	19503.0	0.1	0.3	0.0	1.0

92. ci proportions sedentarismo2

Variable	Obs	Proportion	Std. err.	Binomial exact [95% conf. interval]	
sedentaris~2	19,503	.1410039	.0024921	.1361477	.1459675

93. //[mean of the individual outcomes = "sample mean"]

94.

95. **# Stratum-level descriptive statistics

96. * Load the stratum-level data

97. use "stratum.dta", clear

```

98.
99. * Generate binary indicators for whether each stratum has more than X
100 * individuals
101 generate n100plus = (n >= 100)

102 generate n50plus = (n >= 50)

103 generate n30plus = (n >= 30)

104 generate n20plus = (n >= 20)

105 generate n10plus = (n >= 10)

106 generate nlessthan10 = (n < 10)

107
108 * Tabulate the binary indicators
109 tabulate n100plus

```

n100plus	Freq.	Percent	Cum.
0	6	8.33	8.33
1	66	91.67	100.00
Total	72	100.00	

```
110 tabulate n50plus
```

n50plus	Freq.	Percent	Cum.
1	72	100.00	100.00
Total	72	100.00	

```
111 tabulate n30plus
```

n30plus	Freq.	Percent	Cum.
1	72	100.00	100.00
Total	72	100.00	

```
112 tabulate n20plus
```

n20plus	Freq.	Percent	Cum.
1	72	100.00	100.00
Total	72	100.00	

```
113 tabulate n10plus
```

n10plus	Freq.	Percent	Cum.
1	72	100.00	100.00
Total	72	100.00	

```
114 tabulate nlessthan10
```

nlessthan10	Freq.	Percent	Cum.
0	72	100.00	100.00
Total	72	100.00	

```

115
116 **# Table 3
117 * Load the individual-level data
118 use "individual.dta", clear

119
120 * Model 0
121
122 * Load the estimation results and make them the current (active) results
123 estimates use "m0.ster"

124
125 * Replay the estimation results
126 estimates replay, or baselevels

```

active results

Mixed-effects logistic regression Number of obs = **19,503**
Group variable: **stratum** Number of groups = **72**

Obs per group:

min	=	62
avg	=	270.9
max	=	560

Integration method: **mvaghermite** Integration pts. = **7**

Log likelihood = **-7702.5415** Wald chi2(0) = **.**
Prob > chi2 = **.**

sedentaris~2	Odds	Std. err.	z	P> z	[95% conf. interval]	
_cons	0.14	0.01	-29.10	0.000	0.12	0.16
stratum						
var(_cons)	0.28	0.06			0.19	0.41

Note: Estimates are transformed only in the first equation to odds.
LR test vs. logistic model: chibar2(01) = 461.85 Prob >= chibar2 = **0.0000**

```

127
128 * Calculate the variance partition coefficient (VPC)
129 estat icc

```

Intraclass correlation

Level	ICC	Std. err.	[95% conf. interval]	
stratum	.0778026	.0146322	.0535405	.111761

```

130
131 * Calculate the area under the receiver operating characteristic (ROC) curve
132 * based on fitted linear predictor for the fixed portion of the model only
133 * (only the intercept)
134 roctab sedentaris~2 m0_fixed

```

Obs	ROC area	Std. err.	Asymptotic normal [95% conf. interval]	
19,503	0.5000	0.0000	0.50000	0.50000

135
 136 * Calculate the area under the receiver operating characteristic (ROC) curve
 137 * based on fitted linear predictor (intercept and stratum random effect)
 138 roctab sedentarismo2 m0_total

Obs	ROC area	Std. err.	Asymptotic normal [95% conf. interval]	
19,503	0.6498	0.0056	0.63877	0.66090

139
 140 * Model 1
 141 * Load the estimation results and make them the current (active) results
 142 estimates use "ml.ster"

143
 144 * Replay the estimation results
 145 estimates replay, or baselevels

active results

Mixed-effects logistic regression
 Group variable: **stratum**

Number of obs = 19,503
 Number of groups = 72

Obs per group:
 min = 62
 avg = 270.9
 max = 560

Integration method: **mvaghermite**

Integration pts. = 7

Log likelihood = **-7649.4122**

Wald chi2(7) = 252.56
 Prob > chi2 = 0.0000

sedentarismo2	Odds ratio	Std. err.	z	P> z	[95% conf. interval]	
sexo						
Male	1.00	(base)				
Female	1.91	0.12	10.62	0.000	1.69	2.15
edad_cat3						
6-9	1.00	(base)				
10-12	1.05	0.08	0.65	0.518	0.91	1.22
13-15	1.64	0.12	6.84	0.000	1.42	1.89
clase_2						
Non-Manual W..	1.00	(base)				
Manual Workers	1.62	0.10	7.80	0.000	1.43	1.82
urb_rur						
Rural	1.00	(base)				
Semi-urban	1.34	0.11	3.64	0.000	1.14	1.57
Urban	1.36	0.10	4.02	0.000	1.17	1.58
survey2						
Post	1.00	(base)				
Pre	1.32	0.08	4.39	0.000	1.17	1.49
_cons	0.05	0.00	-30.01	0.000	0.04	0.06
stratum						
var(_cons)	0.03	0.01			0.01	0.06

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

LR test vs. logistic model: chibar2(01) = 14.03 Prob >= chibar2 = 0.0001

```

146
147 * Calculate the variance partition coefficient (VPC)
148 estat icc

```

Residual intraclass correlation

Level	ICC	Std. err.	[95% conf. interval]	
stratum	.0079565	.0032902	.0035305	.0178318

```

149
150 * Calculate the Proportional Change in Variance (PCV) (as a percentage)
151 display %3.1f 100*(m0sigma2 - m1sigma2) / m0sigma2
152 90.5

```

```

152
153 * Calculate the area under the receiver operating characteristic (ROC) curve
154 * based on linear predictor for the fixed portion of the model only (main
155 * effects only)
156 roctab sedentarismo2 m1_fixed

```

Obs	ROC area	Std. err.	Asymptotic normal [95% conf. interval]	
19,503	0.6344	0.0058	0.62307	0.64576

```

157
158 * Calculate the area under the receiver operating characteristic (ROC) curve
159 * based on fitted linear predictor (main effects and interactions)
160 roctab sedentarismo2 m1_total

```

Obs	ROC area	Std. err.	Asymptotic normal [95% conf. interval]	
19,503	0.6451	0.0057	0.63394	0.65631

```

161
162 **# Figure 2 - Predicted Probability
163 * Load the stratum-level data
164 use "stratum.dta", clear

```

```

165
166 * Generate the predicted stratum percentages
167 generate m1_prob_total = 100 * invlogit(m1_total)

```

```

168
169 * Rank the predicted stratum percentages
170 egen m1_prob_total_rank = rank(m1_prob_total)

```

```

171
172 * Generate the lower and upper limits of the approximate 95% confidence
173 * intervals for the predicted stratum percentages
174 generate m1_total_lo = 100 * invlogit(m1_fixed + m1_random ///
> - 1.96 * sqrt(m1_fixed_se^2 + m1_random_se^2))
175 generate m1_total_hi = 100 * invlogit(m1_fixed + m1_random ///
> + 1.96 * sqrt(m1_fixed_se^2 + m1_random_se^2))

```

```

176 // Approximate as the model assumes no sampling covariability between the
177 // regression coefficients and the stratum random effect
178
179 * Plot the caterpillar plot of the predicted stratum means
180
181 * label height
182 generate lab_y = 48
183
184 twoway ///
185 > (rspike ml_total_hi ml_total_lo ml_prob_total_rank, lcolor(gs4)) ///
186 > (scatter ml_prob_total ml_prob_total_rank, mcolor(black) msymbol(smcircle)) ///
187 > (scatter lab_y ml_prob_total_rank, msymbol(none) ///
188 > mlabel(stratum) mlabsize(*0.8) mlabcolor(black) ///
189 > mlabangle(90) mlabposition(12) mlabgap(*1)) ///)
190 > , ///
191 > ytitle("Predicted Percent Sedentarism" " " "Model 2", size(*1.5)) ///
192 > ylabel(0(10)50, angle(horizontal) labsize(*1.5)) ///
193 > yline(10 20 30 40 50 60 70, lwidth(vthin) lcolor(gray)) ///
194 > xtitle("Stratum rank", size(*1.5)) ///
195 > xlabel(0(12)72, labsize(*1.5)) ///
196 > legend(off) ///
197 > scheme(slmono) ///
198 > name(Figure2B, replace) ///
199 > xsize(10)
200
201
202 185
203 186 * Export the graph as a portable network graphics (PNG) file
204 187 graph export "Figure2_predictedprobability_sedentarism.png", replace width(1000)
205 file Figure2_predictedprobability_sedentarism.png saved as PNG format
206
207 188
208 189 * Generate list of 5 highest/lowest predicted stratum percentages (for Table 4)
209 190 sort ml_prob_total_rank
210
211 191 list stratum sexo edad_cat3 clase_2 urb_rur survey2 n ml_prob_total_rank ml_prob_tot
212 > al ml_total_lo ml_total_hi in f/5

```

1.	stratum 11111	sexo Male	edad_cat3 6-9	clase_2 Non-Manual W	urb_rur Rural	survey2 Post	n 75
	ml_prob~k 1	ml_prob~l 4.3875	ml_tot~o 3.084746	ml_tot~i 6.20521			

2.	stratum 12111	sexo Male	edad_cat3 10-12	clase_2 Non-Manual W	urb_rur Rural	survey2 Post	n 81
	ml_prob~k 2	ml_prob~l 4.655903	ml_tot~o 3.27858	ml_tot~i 6.572518			

3.	stratum 11121	sexo Male	edad_cat3 6-9	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Post	n 151
	ml_prob~k 3	ml_prob~l 5.275023	ml_tot~o 3.794945	ml_tot~i 7.288618			

4.	stratum 12112	sexo Male	edad_cat3 10-12	clase_2 Non-Manual W	urb_rur Rural	survey2 Pre	n 191
	ml_prob~k 4	ml_prob~l 5.952586	ml_tot~o 4.328017	ml_tot~i 8.135104			

5.

stratum 12131	sexo Male	edad_c~3 10-12	clase_2 Non-Manual W	urb_rur Urban	survey2 Post	n 251
m1_pro~k 5		m1_pro~l 6.235746		m1_tot~o 4.596965		m1_tot~i 8.407256

```
192 list stratum sexo edad_cat3 clase_2 urb_rur survey2 n m1_prob_total_rank m1_prob_to
> tal m1_total_lo m1_total_hi ///
> in -5/-1
```

68.

stratum 23221	sexo Female	edad_c~3 13-15	clase_2 Manual Worke	urb_rur Semi-urban	survey2 Post	n 157
m1_pro~k 68		m1_pro~l 23.4239		m1_tot~o 18.57581		m1_tot~i 29.08533

69.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
m1_pro~k 69		m1_pro~l 24.17674		m1_tot~o 19.59134		m1_tot~i 29.4424

70.

stratum 23231	sexo Female	edad_c~3 13-15	clase_2 Manual Worke	urb_rur Urban	survey2 Post	n 249
m1_pro~k 70		m1_pro~l 26.08607		m1_tot~o 21.30369		m1_tot~i 31.51213

71.

stratum 23222	sexo Female	edad_c~3 13-15	clase_2 Manual Worke	urb_rur Semi-urban	survey2 Pre	n 351
m1_pro~k 71		m1_pro~l 31.16492		m1_tot~o 26.2306		m1_tot~i 36.56733

72.

stratum 23232	sexo Female	edad_c~3 13-15	clase_2 Manual Worke	urb_rur Urban	survey2 Pre	n 503
m1_pro~k 72		m1_pro~l 31.64188		m1_tot~o 27.03463		m1_tot~i 36.64002

```
193
194 **# Figure 3 - Residual Effects
195 * Load the stratum-level data
196 use "stratum.dta", clear

197
198 * Replaces each observation in the dataset with 1000 copies of the observation
199 expand 1000
    (71,928 observations created)
```

```

200 // We do this as for this plot we follow a simulation-based approach to
201 // calculating the limits of the approximate 95% confidence intervals of our
202 // predictions. This involves simulating 1000 values for each predicted value.
203 //
204 // Approximate as the model assumes no sampling covariability between the
205 // regression coefficients and the stratum random effect
206
207 * Generate the approximate standard error for the linear predictor
208 generate ml_total_se = sqrt(ml_fixed_se^2 + ml_random_se^2)
209
210 * Specify initial value of random-number seed
211 set seed 354612
212
213 * Generate the predicted stratum percentages based on the regression
214 * coefficients and the predicted stratum random effect and factoring in
215 * prediction uncertainty
216 generate ml_prob_total_sim = 100 * invlogit(ml_fixed + ml_random + rnormal(0, ml_tot
  > al_se))
217
218 * Generate the predicted stratum percentages ignoring the predicted stratum
219 * effect
220 generate ml_prob_fixed_sim = 100 * invlogit(ml_fixed)
221
222 * Generate the difference in the predicted stratum percentages due to the
223 * predicted stratum effect
224 generate ml_diff = ml_prob_total_sim - ml_prob_fixed_sim
225
226 * Generate the lower and upper limits of the approximate 95% confidence
227 * intervals for the difference in predicted stratum percentages due to
228 * interaction
229 bysort stratum: egen ml_diff_lo = pctlile(ml_diff), p(2.5)
230 bysort stratum: egen ml_diff_hi = pctlile(ml_diff), p(97.5)
231
232 * — SAVE original data with stratum-level vars BEFORE collapsing —
233 preserve
234
235 * Convert the data into a stratum-level dataset
236 collapse (mean) ml_diff, by(stratum ml_diff_lo ml_diff_hi)
237
238 * Rank the predicted stratum percentage differences
239 egen ml_diff_rank = rank(ml_diff)
240
241 * Save stratum-level results
242 save "strata2.dta", replace
  file strata2.dta saved
243
244 * Re-plot the caterpillar of the predicted stratum percentage differences
245 generate lab_y = 14

```

```

246
247 twoway ///
>      (rspike ml_diff_hi ml_diff_lo ml_diff_rank, lcolor(gs4)) ///
>      (scatter ml_diff ml_diff_rank, mcolor(black) msymbol(smcircle)) ///
>      (scatter lab_y ml_diff_rank, msymbol(none)) ///
>      mlabel(stratum) mlabsize(*0.8) mlabcolor(black) ///
>      mlabangle(90) mlabposition(12) mlabgap(*1)) ///
>      , ///
>      ytitle("Difference in Predicted Percent" ///
>      " " "Sedentarism due to Interactions", size(*1.5)) ///
>      yline(0) ///
>      yline(-10 -5 5 10 15, lcolor(gray)) ///
>      ylabel(-10(5)15, angle(horizontal) labsize(*1.5)) ///
>      xtitle("Stratum rank", size(*1.5)) ///
>      xlabel(0(12)72, labsize(*1.5)) ///
>      legend(off) ///
>      scheme(slmono) ///
>      name(Figure3C, replace) ///
>      xsize(10)

248
249 * Export the graph as a portable network graphics (PNG) file
250 graph export "Figure 3_residualeffects.png", replace width(1000)
    file Figure 3_residualeffects.png saved as PNG format

251
252 * — RESTORE original data and merge in the new stratum-level vars —
253 restore

254
255 * Merge in the stratum-level results
256 merge m:1 stratum using "strata2.dta"
    (label stratum already defined)

```

Result	Number of obs	
Not matched	0	
Matched	72,000	(_merge==3)

```

257 drop _merge

```

```

258
259 * Generate list of 5 highest/lowest residual effects strata
260
261 sort ml_diff_rank

```

```

262 list stratum sexo edad_cat3 clase_2 urb_rur survey2 n ml_diff_rank ml_diff ml_diff_1
> o ml_diff_hi in f/5

```

1.	stratum 21132	sexo Female	edad_cat3 6-9	clase_2 Non-Manual W	urb_rur Urban	survey2 Pre	n 514
	ml_diff~k 1	ml_diff -3.923442		ml_diff~o -4.989115		ml_diff~i .4280205	
2.	stratum 21132	sexo Female	edad_cat3 6-9	clase_2 Non-Manual W	urb_rur Urban	survey2 Pre	n 514
	ml_diff~k 1	ml_diff -.4263048		ml_diff~o -4.989115		ml_diff~i .4280205	

3.

stratum 21132	sexo Female	edad_c~3 6-9	clase_2 Non-Manual W	urb_rur Urban	survey2 Pre	n 514
ml_dif~k 1		ml_diff -3.213659	ml_diff~o -4.989115		ml_dif~i .4280205	

4.

stratum 21132	sexo Female	edad_c~3 6-9	clase_2 Non-Manual W	urb_rur Urban	survey2 Pre	n 514
ml_dif~k 1		ml_diff -2.496414	ml_diff~o -4.989115		ml_dif~i .4280205	

5.

stratum 21132	sexo Female	edad_c~3 6-9	clase_2 Non-Manual W	urb_rur Urban	survey2 Pre	n 514
ml_dif~k 1		ml_diff -3.072948	ml_diff~o -4.989115		ml_dif~i .4280205	

```
263 list stratum sexo edad_cat3 clase_2 urb_rur survey2 n ml_diff_rank ml_diff ml_diff_l
> o ml_diff_hi ///
> in -5/-1
```

71996.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
ml_dif~k 72		ml_diff 4.462572	ml_diff~o -.7405663		ml_dif~i 9.239945	

71997.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
ml_dif~k 72		ml_diff 3.457743	ml_diff~o -.7405663		ml_dif~i 9.239945	

71998.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
ml_dif~k 72		ml_diff .2092018	ml_diff~o -.7405663		ml_dif~i 9.239945	

71999.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
ml_dif~k 72		ml_diff 3.825914	ml_diff~o -.7405663		ml_dif~i 9.239945	

72000.

stratum 23122	sexo Female	edad_c~3 13-15	clase_2 Non-Manual W	urb_rur Semi-urban	survey2 Pre	n 246
ml_dif~k 72		ml_diff 3.97044	ml_diff~o -.7405663		ml_dif~i 9.239945	

```
264
265 **# Close log file
266 capture log close
```